



TYPE: Article

Title: Spooky Cognition: Quantum Entanglement, Ephaptic Coupling, and the Transcendental Frontiers of Consciousness Studies

Author(s): Carl A. Raschke

Source: Journal for Cultural and Religious Theory (JCRT)

Published by: Whitestone Publications

Stable URL: <https://jcrt.org/archives/25.2/raschke/>

Copyright © held by the author(s). All rights reserved. This text may be used and shared in accordance with the fair-use provisions of U.S. copyright law. Any use of this text in other ways requires the consent of the author and the publisher, the *Journal for Cultural and Religious Theory*, and must cite publication in this journal.

<https://jcrt.org/copyright/>



CARL RASCHKE

University of Denver CO USA

SPOOKY COGNITION: QUANTUM ENTANGLEMENT, EPAPTIC COUPLING, AND THE TRANSCENDENTAL
FRONTIERS OF CONSCIOUSNESS STUDIES

Abstract

This article explores the convergence of quantum physics, neuroscience, and philosophy of religion in understanding consciousness and spiritual experience. Beginning with quantum entanglement and Einstein's characterization of “spooky action at a distance”, the article traces an analogy to recent discoveries in neuroscience, particularly ephaptic coupling – the non-synaptic, field-based communication between neurons. Drawing on David Chalmers' 'hard problem' of consciousness and critiquing physicalist reductionism, the article argues that contemporary neuroscience, when properly integrated with theoretical physics and phenomenological philosophy, opens unprecedented avenues for investigating transcendent experiences. The work of Andrew Newberg in neurotheology serves as a bridge between empirical brain imaging and theological inquiry, demonstrating how meditation, prayer, and mystical states correlate with specific patterns of neural activity. By invoking Kant's transcendental philosophy and post-structural semiotics, the article proposes a framework for 'transcendental semiotics' that moves beyond the immanence/transcendence dichotomy, suggesting that quantum-level information processing and electromagnetic field effects in the brain may provide a scientific basis for understanding religious and spiritual phenomena without resorting to reductionism.

Introduction: From Quantum Spookiness to Neural Mysteries

During the 1930s quantum physics confronted a phenomenon that profoundly challenged classical notions of causality and locality – “non-locality”, or what came to be colloquially known as “action at a distance”. This phenomenon, carefully documented in experiments and predicted by quantum theory itself, consists in the capacity whereby one object can influence another object at any distance without identifiable physical contact or interaction. Action at a distance implies that a change or effect in one entity or place can instantaneously occasion a change in another entity or place, no matter how far apart. The discovery of non-locality upended classical notions of causality and locality and baffled both Albert Einstein and his University of Berlin colleague Erwin Schrödinger. The phenomenon of non-locality eventually came to be explained through the theory of “quantum entanglement”,¹ which posits that the quantum state of any given particle cannot be described independently of the state of others, even when the particles are separated by large distances.² Whenever particles become entangled, their properties become correlated with each other to the extent that the measurement of the state of one particle instantaneously determines the state of the other, regardless of the span separating them. The correlation of the particles cannot be accounted for by signals passing between them. On the contrary, the entangled system of particles simulates one single, unified quantum state.

Einstein termed the notion of quantum entanglement “spooky action at a distance”, mainly because it seemed to imply instantaneous communication between the distant particles.³ Einstein's theory

¹John Stewart Bell, “On the Einstein Podolsky Rosen Paradox” ,*Physics Physique Физика* 1, no. 3 (1964): 195-200. The phenomenon of quantum non-locality challenged the principle of locality fundamental to classical physics and special relativity.

²For a comprehensive overview of quantum entanglement theory, see Alain Aspect, 'Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A New Violation of Bell's Inequalities,' *Physical Review Letters* 49, no. 2 (1982): 91-94.

³Albert Einstein, Boris Podolsky, and Nathan Rosen, 'Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?' *Physical Review* 47, no. 10 (1935): 777-780. Einstein's discomfort with quantum mechanics' implications for locality and causality is well documented in his correspondence

of special relativity holds that nothing can travel faster than the speed of light, and henceforth no signal or information can propagate beyond this limit. Entangled particles, however, appear to affect each other's state immediately upon measurement. Along with physicists Boris Podolsky and Nathan Rosen, Einstein set forth the so-called EPR paradox (as an acronym for their last names) in 1935 to suggest that quantum mechanics posed a "paradox" and therefore proved 'incomplete'.⁴ They contended there must be certain 'hidden variables' conditioning the states of particles, thereby re-instating the 'common sense' assumption that correlation can be independent of causation.

Given the epochal "paradigm shift", if we may use Thomas Kuhn's celebrated but overdetermined phrasing, that Einstein's theory of special relativity had recently unleashed in theoretical physics, this position held its own until John Bell demonstrated in the mid-1960s that, in fact, there were no hidden variables.⁵ Critical experiments in the 1970s and 1980s, especially those conducted by Alain Aspect and his team, confirmed that quantum entanglement is real and a fundamental principle of physics.⁶ More recently, advanced modeling and research in the physical sciences has turned its attention certain highly refined classical theories of gravity, which boggle the mind in ways even Einstein could not have imagined, have laid the groundwork for eventually resolving the EPR paradox in the measure that it is longer can be seen as a "paradox" per se. These advances have also led to opportunities for major strides in the development of what is known as "quantum computing," which promises to be the next epochal technological revolution that could surpass artificial intelligence.

Yet what does any of the foregoing have to do with the phenomenon of religion, or our more comprehensive understanding of what religion is? Everything and nothing at the same time. Even though theoretical physics in the past half century has been more amenable to entertaining theological or religious questions than the other "hard sciences", it is the fast-evolving field of neuroscience (colloquially known as 'brain science') that can in principle at least address how spiritual conundra can be investigated in a responsible and credible manner. Unfortunately, until the turn of the millennium, neuroscience was trapped in an oversimplified "physicalist" reference frame of its own making.

The Challenge to Physicalism: Chalmers and the Hard Problem

Physicalism, an offshoot of so-called "mind versus matter" debates in philosophy that have exerted sway ever since Descartes, consists in the view that everything that exists in nature is material and can be explained in terms of tangible entities or material processes.⁷ It maintains that any kind of occurrence, whether "objective" or "subjective", including mental and affective data, are physical in nature. Physicalism also holds as a dogma that mind itself is reducible to observable neural signatures of brain activity, which in turn require empirical or quantitative assessment through external instruments such as an encephalograph. Physicalism, therefore, disallows for any force or factor that transcends the range of easily detectable phenomena and exhibits some kind of "causal" connection with them. Physicalism in the neurosciences is tantamount to what is known as "local realism" in theoretical physics.

Australian philosopher David Chalmers in his 1996 book *The Conscious Mind* was the first to seriously challenge the long-regnant physicalist orthodoxy of the Anglo-American philosophical

⁴Ibid. The EPR paper argued that quantum mechanics must be supplemented by "hidden variables" to provide a complete description of physical reality.

⁵John Bell, "On the Einstein Podolsky Rosen Paradox." Bell's theorem demonstrated that no local hidden-variable theory could reproduce all the predictions of quantum mechanics.

⁶Aspect's experiments definitively ruled out local hidden-variable theories and confirmed the reality of quantum entanglement. See Alain Aspect, Jean Dalibard, and Gérard Roger. "Experimental Test of Bell's Inequalities Using Time-varying Analyzers." *Physical review letters* 49, no. 25 (1982): 180. See also Aspect, Alain. "Bell's theorem: the Naive View of an Experimentalist." In *Quantum [Un] speakables: From Bell to Quantum Information*, 119-153. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012.

⁷For a thorough analysis of physicalism in philosophy of mind, see Jaegwon Kim, *Physicalism, or Something Near Enough* (Princeton: Princeton University Press, 2005).

establishment with what he dubbed the 'hard problem' critique of physicalism.⁸ The "hard problem", according to Chalmers, amounts to the formidable challenge of explaining how and why physical states, events, or processes in the brain can generate what are called "qualia", or conscious feelings and sensations. In comparison with the "easy problems" of consciousness – that is, accounts of cognitive operations and perceptible behavior that manifest physically – the "hard problem" derives from a persistent explanatory gap between brain functions and subjective experience.⁹ As Chalmer writes:

It still seems utterly mysterious that the causation of behavior should be accompanied by a subjective inner life. We have good reason to believe that consciousness arises from physical systems such as brains, but we have little idea how it arises, or why it exists at all. How could a physical system such as a brain also be an experiencer? Why should there be something it is like to be such a system? Present-day scientific theories hardly touch the really difficult questions about consciousness. We do not just lack a detailed theory; we are entirely in the dark about how consciousness fits into the natural order.¹⁰

Stanislas Dehaene, however, remains skeptical about Chalmers' "hard problem". He argues that Chalmers merely "swapped the labels" of "hard and easy". He writes:

It is the 'easy' problem that is hard, while the hard problem just seems hard because it engages ill-defined intuitions. Once our intuition is educated by cognitive neuroscience and computer simulations, Chalmers's hard problem will evaporate. The hypothetical concept of qualia, pure mental experience detached from any information-processing role, will be viewed as a peculiar idea of the prescientific era, much like vitalism – the misguided nineteenth-century thought that, however much detail we gather about the chemical mechanisms of living organisms, we will never account for the unique qualities of life. Modern molecular biology shattered this belief, by showing how the molecular machinery inside our cells forms a self-reproducing automaton. Likewise, the science of consciousness will keep eating away at the hard problem until it vanishes.¹¹

Transcendental Philosophy and the Linguistic Turn

The seesaw battle in philosophy throughout the modern era, commonly portrayed as physicalism (or materialism) versus idealism, or monism versus dualism, misses an obvious but habitually overlooked development from the early twentieth century that altered the very terms of the controversy itself. It is the recognition, pioneered by Wittgenstein but refined by post-structuralism, that every squabble over what is "real" and what is not depends on the semiotic interventions of human beings.¹² As Heidegger famously put it, *Die Sprache spricht* ("language speaks"). Yet how does it speak? Is it genuinely a "mirror of nature", as Richard Rorty in his groundbreaking book of 1980 asked, or is it something else?¹³

One must exercise caution, however, in invoking the mirror metaphor, regardless of whether the aim is to uphold or oppose physicalism. Kant himself, in effect, realigned the conversation with his key insight that phenomena and noumena in the empirical sciences are totally separable from

⁸For a thorough analysis of physicalism in philosophy of mind, see Jaegwon Kim, *Physicalism, or Something Near Enough* (Princeton: Princeton University Press, 2005).

⁹David J. Chalmers, 'Facing Up to the Problem of Consciousness,' *Journal of Consciousness Studies* 2, no. 3 (1995): 200-219. The distinction between "easy" and "hard" problems of consciousness has become foundational to contemporary consciousness studies.

¹⁰*Ibid.* xi.

¹¹Stanislas Dehaene, *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts* (New York: Viking, 2014), 262.

¹²*Ibid.*

¹³See Richard Rorty, *Philosophy and the Mirror of Nature* (Princeton: Princeton University Press, 1978).

each other (though of course not in the moral sense). They are, in fact, what Peter Strawson once designated as “limit concepts”.¹⁴ However, pure sense data are not irreducible either. They only acquire content when they are processed through the operations of what Kant referred to as a “transcendental logic”. The word “transcendental” refers to the structures, conditions, or *a priori* precepts that render human knowledge and experience possible in the first place.¹⁵ French post-structuralism, beginning with Ferdinand de Saussure, discovered that transcendental logic in truth is the same as the administration of grammatical rules in the human production of sentences, texts, and conversations.¹⁶ He termed this particular “logic” *langue* – the grammar or language system itself – as opposed to *parole*, i.e., discursive elaborations. *Langue* is something we inherit and absorb culturally. *Parole* is its performative application by individuals. For Saussure, therefore, *langue* corresponds in multiple respects to Kant’s ‘categories of understanding.’ The so-called “linguistic turn” in late modern philosophy was at bottom a reinvention of Kantianism.

Anglo-American analytical philosophy has been obsessed all along with predicative logic. But French post-structuralists as the scions of Saussure have introduced the term *signifiante* – coined by Saussure and more commonly rendered “signification” in English – which implies an apparatus for conveying meaning in a broader sense.¹⁷ It also denotes how signs and symbols result in the formation of concepts and objects. The search for *signifiante* is crucial to fields like semiotics and hermeneutics, which investigate how meaning is configured and disseminated. Technical applications of *signifiante* in semiotics and linguistics uncover a sprawling ensemble of structural and associative ways in which signification works in both language and thought.

Ephaptic Coupling: Neural Action at a Distance

Let us now consider a brief illustration of these new “frontiers” in cognition that neuroscience has opened up. We referenced earlier how quantum theory disrupted the mechanistic paradigm of classical, or Newtonian, physics with such force that even Einstein was not willing to concede some of the most extreme implications of the scientific revolution it ignited. What Einstein dubbed “spooky” interactions have distinct analogues in recent specific discoveries that can only be explicated by pairing theoretical physics with brain science.

One somewhat esoteric instance is so-called “ephaptic coupling”. Ephaptic coupling constitutes a hypothesized physical process whereby local electric fields generated from neuronal activity in the brain influence and modulate nearby neuronal firings without any evidence of a synaptic interface.¹⁸ In short, ephaptic coupling is the neuroscientific version of “action at a distance.” Ephaptic coupling has been observed through certain experiments at the California Institute of Technology yielding somewhat ambiguous yet potentially paradigm-busting data. Empirical findings show that even very weak extracellular fields can alter neuronal firing and augment synchrony between different sites in the brain.¹⁹ Such coupling significantly impacts the computational strategies of the brain, according to researcher Christof Koch in collaboration with Costas Anastassiou and others.²⁰

As a publicity release from Caltech puts it, these effects can become pronounced during conditions like epileptic seizures, where much stronger fields are generated. The release states:

¹⁴P.F. Strawson, *The Bounds of Sense: An Essay on Kant's Critique of Pure Reason* (London: Methuen, 1966).

¹⁵Immanuel Kant, *Critique of Pure Reason*, trans. Norman Kemp Smith (New York: St. Martin's Press, 1965). For Kant, transcendental conditions are not empirical but are the necessary preconditions for any empirical knowledge whatsoever.

¹⁶Ferdinand de Saussure, *Course in General Linguistics*, trans. Wade Baskin (New York: Philosophical Library, 1959).

¹⁷Julia Kristeva, 'The System and the Speaking Subject,' *Times Literary Supplement*, October 12, 1973. Kristeva elaborates on Saussure's concept of *signifiante* as distinct from mere signification.

¹⁸Costas A. Anastassiou and Christof Koch, “Ephaptic Coupling to Endogenous Electric Field Activity: Why Bother?” *Current Opinion in Neurobiology* 31 (2015): 95-103.

¹⁹*Ibid.*, 217-223.

²⁰*Ibid.* Koch and Anastassiou's research demonstrates that extracellular fields can causally affect neural function under physiological conditions.

The brain—awake and sleeping—is awash in electrical activity, and not just from the individual pings of single neurons communicating with each other. In fact, the brain is enveloped in countless overlapping electric fields, generated by the neural circuits of scores of communicating neurons. The fields were once thought to be an “epiphenomenon” similar to the sound the heart makes—which is useful to the cardiologist diagnosing a faulty heartbeat, but doesn't serve any purpose to the body.²¹

A summary of available neuroscientific literature shows that ephaptic coupling is increasingly recognized as a complementary form of neural and cardiac electrical interaction alongside synaptic operations.²² Ephaptic coupling plays a key role in calibrating neural synchrony and circuit conductivity, particularly in the context of certain pathologies. Its subtle but impactful contributions to brain function include memory network formation, oscillatory propagation, and seizure modulation. Viewed from the standpoint of physics, ephaptic coupling takes place through the synergy of electric fields arising out of interactions between adjacent neuronal membrane potentials. This phenomenon itself is a consequence of the influence of extracellular electric fields that are produced by ionic currents streaming through active neurons. These fields condition the membrane voltage of nearby neurons in their packed extracellular microenvironment. The fields themselves depolarize or hyperpolarize neurons sufficiently to excite them and increase their firing probability. The precise geometry of neuronal spacing decisively shapes the intensity and distribution of these fields and hence the strength of ephaptic interactions.

Other theoretical and applied studies make clear that exposure by the brain to extremely low-frequency (ELF) electromagnetic fields originating in the wider environment are capable of modulating neuronal plasticity, neurogenesis, and synaptic interaction—what are known as “action potentials”. These fields can synchronize neural firings and configure both cognition and behavior.²³ Similarly, they can only create the monitored effects which the neuroscientific literature has cumulatively begun documenting. The information that modulates these external fields obviously impacts the internal changes in the brain that encephalography and other measuring devices register. The information encoded in these fields can only be inferred from the electric signatures in the brain which the mind correlates and construes as “subjective experience”. Chalmers' “hard problem”, therefore, refuses to go away. But even a naïve version of physicalism has far greater difficulties in understanding these intricate cerebral transits by which information is encoded and decoded. A committed physicalist is like an obdurate exponent of Newtonian mechanics who clings to the belief that the sole viable metaphor for cosmological inquiry is the collision of billiard balls on a table in three-dimensional space. One cannot explain quasars in the same manner that Newton accounted for gravity as in the apocryphal tale of an apple dropping on his head from a tree.

“Neurotheology”: Andrew Newberg and the Empirical Study of Spiritual Experience

Given the quite genuine possibility of information conveyance through weak bio-electrical fields resulting from synaptic voltage differentials, we must ask ourselves what are the tacit “theological” ramifications of such experiments? The spatial and temporal coherence of ambient electromagnetic fields and their interaction with endogenous brain activity determines the efficacy of such information transmittal—that is, the so-called ‘signal to noise’ ratio. There is compelling evidence that the brain integrates these external and internal signals in what amounts to electric field-based communication between the brain and the larger universe.

²¹ California Institute of Technology, “Neurobiologists Find that Weak Electrical Fields in the Brain Help Neurons Fire Together”, press release, January 16, 2011, <https://www.caltech.edu/about/news/neurobiologists-find-weak-electrical-fields-brain-help-neurons-fire-together-1671>. Accessed Oct. 24, 2025.

²² D.A. Pinotsis and E.K. Miller, “In Vivo Ephaptic Coupling Allows Memory Network Formation”, *Cerebral Cortex* 33, no. 17 (2023): 9877-9895.

²³ Ibid., 9877-9895.

The emergent partnership between physics and neuroscience thus opens the gates wide for an even more fruitful collaboration with theological studies in a setting that heretofore has been regarded as unimaginable. We are talking not merely about “spooky” types of cognition, but what might be described as an extremely generative “transcendental” perspective (in Kant's sense) that also involves philosophy tasked with asking the biggest and most fundamental questions. Contra Kant, however, such a radical epistemological stance would enable a new “transcendental semiotics” to probe into the noumenal or transcendent, and vice-versa. There are countless thought experiments, let alone actual experimental undertakings, that exemplify how we might go about this complex venture. Noteworthy in the latter case is the work of Andrew Newberg. Newberg is a prominent neuroscientist whose research has focused on examining how a range of cognitive states, especially those connected to meditation, prayer, ritual performance, and trance, correspond to specific patterns of brain activity.²⁴ Newberg coined the expression “neurotheology”.²⁵

Newberg has deployed sophisticated brain imaging techniques, such as single-photon emission computed tomography (SPECT) and functional MRI, to study the brains of Buddhist meditators, Franciscan nuns, and other individuals engaged in spiritual practices.²⁶ He has identified neurological mechanisms underlying religious experiences, such as feelings of transcendence, oneness, and altered sensations of time and self. One of his experiments has demonstrated that decreased activity in the parietal lobes during deep meditation correlates with a loss of the routine sense of spatial parameters, fostering mystical experiences.²⁷ His research also probes into the ways in which spiritual deliberations influence human health, both mental and physical. Newberg has consistently stressed the importance of preserving scientific rigor while respecting the integrity of religious belief. His contention is that neurotheology can supply valuable insights into the biological fundamentals of spiritual approaches minus the familiar feint of physicalism and methodological reductionism.

However, Newberg for the most part steers clear of the blended discipline of biophysics, and his overall approach is much more experimental than theoretical. It is in the theoretical arena where the philosophy of religion, or “philosophical theology”, can be exceptionally helpful.²⁷ The theological task comes down to posing far more challenging questions and pushing the outer limits of what in the parlance of *Gedankenexperiment* or *Gedankenversuch* is known as the exercise of the “secondary imagination”. The secondary imagination allows us to envision both outlandish possibilities and conceptual affinities that tend to be discarded or even censored in much modern and postmodern discourse.

Beyond Immanence: Signals of Transcendence in the Age of New Materialism

Broadly speaking, we have in mind here the inbred prejudice in contemporary philosophy, if not the entirety of academia, toward 'immanence' in setting the terms for any legitimate discussion of spiritual experience. Immanence always seems to trump any effort to decipher what the late sociologist Peter Berger characterized as “signals of transcendence”.²⁸ The popularity of the self-proclaimed “new materialism”, all the rage of late in the Anglosphere of erudite conversation, is a case in point. The new materialism leverages quantum theory to update and refine what is on the face of it the old-school materialism of the nineteenth century.

As I myself wrote and published back in the late 1970s and 1980s, science itself — especially quantum

²⁴ Andrew B. Newberg, *Principles of Neurotheology* (Surrey, UK: Ashgate Publishing, 2010).

²⁵ Ibid., 3-12. Newberg defines neurotheology as “a multidisciplinary field of scholarship that seeks to understand the relationship specifically between the brain and theology and, more broadly, between the mind and religion”.

²⁶ Andrew B. Newberg et al., “The Measurement of Regional Cerebral Blood Flow During the Complex Cognitive Task of Meditation: A Preliminary SPECT Study”, *Psychiatry Research: Neuroimaging* 106, no. 2 (2001): 113-122.

²⁷ Andrew B. Newberg and Eugene G. d'Aquili, *Why God Won't Go Away: Brain Science and the Biology of Belief* (New York: Ballantine Books, 2001), 145-162.

²⁸ Peter L. Berger, *A Rumor of Angels: Modern Society and the Rediscovery of the Supernatural* (Garden City, NY: Doubleday, 1969).

theory — has “raided religion's patent” on the “supernatural”.²⁹ And now it is happening even with outsize strides in information technology — i.e., applied cognitive science — with such still indeterminate advances in chip design that render imminent for the first time a still hazy and still marginally apprehensible far horizon of “quantum computing”, one that could take machine intelligence light years beyond AI and to a hitherto unattained plateau. The conceptual architecture for engineering quantum computing, I understand, mirrors the broader “deep framing: for specific biophysical and “neurotheological” breakthroughs that are presently underway. I am convinced machines will never outpace the development of human intelligence — the fanciful “singularity” AI gurus have nattered about for over a generation — simply because the latter requires feedback from the former in the way that the astounding complexity, parallel processing capabilities, and illimitable signal input sensitivity of the brain empowers. Yet if human intelligence itself does not fall behind that of computers, the geometrically accelerating progress in machine learning will indeed narrow the gap consequentially, especially if the growth rate of median cognitive development falters. The latter is inevitable if global educational institutions do not quickly reform themselves to meet these extraordinary challenges.

Conclusion: Toward a Transcendental Semiotics of Consciousness

What the new generation of neuroscience seems to be haphazardly stumbling upon through standard investigative techniques forces demands a thorough rethink not only of the traditional mind-body template within which theoretical problems themselves are proffered, but even more importantly the lazy distinction philosophers and even theologians all too frequently and reflexively draw between “immanence” and “transcendence”. As Hamlet confesses to his friend in Shakespeare's celebrated play, “there are more things in heaven and earth, Horatio, than are dreamt of in your philosophy”.³⁰ And what we dream may soon become what we straightaway experience in anomalous kingdoms of cognition that were previously thought to be little more than 'metaphysical' flashes of fantasy impounded forever in a cloud cuckooland.

In light of the current state of advanced research in physics, the idea of information that is shared between interlocking dimensions of spacetime is largely speculative, yet continues to be highly plausible within those “speculative” frameworks such as higher-dimensional physics, string theory, and cutting-edge quantum theory. Imperceptible dimensions beyond the familiar quadruple space-time manifold are conjectured to have physical reality, often compactified or concealed at microscopic levels. The now very much accepted framework of 'entangled' particles has led some theories to offer surmises concerning hidden spaces where quantum information 'resides,' or is shared instantaneously. If the quantum version of field theory in higher-order physics allows for the non-causal, non-local, and non-linear correlation of infinitesimal entities, as well as between both seen and unseen dimensions, and their associated packets of information, why should it be disallowed in brain science? The answer is self-evident. It should not be.

The convergence of quantum physics, neuroscience, and theological inquiry represents not a reduction of the spiritual to the material, nor an inflation of the material into the spiritual, but rather a recognition that the traditional categories themselves may be inadequate to the phenomena we are now capable of investigating. Ephaptic coupling, quantum entanglement, and the neurological correlates of mystical experience all point toward a universe more interconnected, more information-rich, and more mysterious than either reductive physicalism or naïve dualism can adequately describe.

This article has argued for what might be called a “transcendental semiotics” — a framework that acknowledges both the Kantian insight that all experience is mediated through structures

²⁹See Carl Raschke, “From God to Infinity, or How Science Raided Religion's Patent on Mystery.” *Zygon* 17, no. 3 (1982): 227-242.

³⁰William Shakespeare, *Hamlet*, Act 1, Scene 5.

of understanding, and the post-structuralist recognition that these structures are fundamentally linguistic and semiotic in nature. Yet unlike both Kant and the post-structuralists, this framework remains open to the possibility that through these very structures of mediation, we may indeed gain access to dimensions of reality that transcend the purely immanent. The brain's capacity for field-based information processing, its sensitivity to quantum-level phenomena, and its remarkable ability to generate experiences of transcendence and unity suggest that consciousness may be both more deeply embedded in the physical universe and more capable of accessing its hidden dimensions than we have previously imagined.

Bibliography

- Anastassiou, Costas A., and Christof Koch. 'Ephaptic Coupling to Endogenous Electric Field Activity: Why Bother?' *Current Opinion in Neurobiology* 31 (2015): 95-103.
- Anastassiou, Costas A., Rodrigo Perin, Henry Markram, and Christof Koch. 'Ephaptic Coupling of Cortical Neurons.' *Nature Neuroscience* 14, no. 2 (2011): 217-223.
- Aspect, Alain. 'Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A New Violation of Bell's Inequalities.' *Physical Review Letters* 49, no. 2 (1982): 91-94.
- Bell, John Stewart. 'On the Einstein Podolsky Rosen Paradox.' *Physics Physique Φιζικα* 1, no. 3 (1964): 195-200.
- Berger, Peter L. *A Rumor of Angels: Modern Society and the Rediscovery of the Supernatural*. Garden City, NY: Doubleday, 1969.
- Chalmers, David J. *The Conscious Mind: In Search of a Fundamental Theory*. New York: Oxford University Press, 1996.
- Chalmers, David J. 'Facing Up to the Problem of Consciousness.' *Journal of Consciousness Studies* 2, no. 3 (1995): 200-219.
- Dehaene, Stanislas. *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts*. New York: Viking, 2014.
- Derrida, Jacques. *Of Grammatology*. Translated by Gayatri Chakravorty Spivak. Baltimore: Johns Hopkins University Press, 1976.
- Einstein, Albert, Boris Podolsky, and Nathan Rosen. 'Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?' *Physical Review* 47, no. 10 (1935): 777-780.
- Kant, Immanuel. *Critique of Pure Reason*. Translated by Norman Kemp Smith. New York: St. Martin's Press, 1965.
- Kim, Jaegwon. *Physicalism, or Something Near Enough*. Princeton: Princeton University Press, 2005.
- Kristeva, Julia. 'The System and the Speaking Subject.' *Times Literary Supplement*, October 12, 1973.
- Newberg, Andrew B. *Principles of Neurotheology*. Surrey, UK: Ashgate Publishing, 2010.
- Newberg, Andrew B., and Eugene G. d'Aquili. *Why God Won't Go Away: Brain Science and the Biology of Belief*. New York: Ballantine Books, 2001.
- Newberg, Andrew B., A. Alavi, M. Baime, M. Pourdehnad, J. Santanna, and Eugene G. d'Aquili. 'The Measurement of Regional Cerebral Blood Flow During the Complex Cognitive Task of Meditation: A Preliminary SPECT Study.' *Psychiatry Research: Neuroimaging* 106, no. 2 (2001): 113-122.
- Pinotsis, D.A., and E.K. Miller. 'In Vivo Ephaptic Coupling Allows Memory Network Formation.' *Cerebral Cortex* 33, no. 17 (2023): 9877-9895.

Rorty, Richard. *Philosophy and the Mirror of Nature*. Princeton: Princeton University Press, 1979.

Ruffini, A., M. Fox, O. Ripolles, P.C. Miranda, and M.A. Pascual-Leone. 'Transcranial Current Brain Stimulation (tCS): Models and Technologies.' *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 21, no. 3 (2013): 333-345.

Saussure, Ferdinand de. *Course in General Linguistics*. Translated by Wade Baskin. New York: Philosophical Library, 1959.

Shakespeare, William. *Hamlet*.

Strawson, P.F. *The Bounds of Sense: An Essay on Kant's Critique of Pure Reason*. London: Methuen, 1966.

Wittgenstein, Ludwig. *Philosophical Investigations*. Translated by G.E.M. Anscombe. Oxford: Blackwell, 1953.